



Technical Service Bulletin: TSB220196

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Electric Starting Motor Changes

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- KV38 (All Versions)
- KV50 (All Versions)
- QSK38 (All Versions)
- QSK50 (All Versions)

Description of Change

A new starting motor is available.

Reason for Change

Production has ceased on all MS7 and M125 starting motor and the M128 starting motor is the replacement.

Service Instructions

Parts and Compatibility

- If using a single (quantity 1) MS7 or M125 starting motor on an engine. The M128 starting motor is a direct replacement.
- If using dual (quantity 2) MS7 or M125 starting motor on an engine. Then both starting motors **must** be replaced with M128 starting motor.
- If using triple (quantity 3) starting motor. All 3 starting motor **must** be replaced with M128 starting motor.
- Table 3 shows starting motor part number supersession and all new part numbers.
- New brackets have been released to mount the magnetic switches to the M128 starting motor and are listed in Table 3.

Installation

- The orientation of the starting motor when fitting to the flywheel housing is dictated by the flat side of the mounting flange. The flat side of the mounting flange is the side that **must** be

oriented to the side face of the cylinder block for correct starting motor orientation. Figure 1 shows the flat face of the starting motor mounting flange.

- The orientation of the starting motor will also dictate the magnetic switch bracket to be used.
- Figure 2 shows the correct mounting bracket depending on starting motor orientation.
- Table 2 communicates the correct magnetic switch bracket part number.
- Figure 3 shows the exploded starting motor assembly detailing the magnetic switch and is applicable for both orientations of mounting bracket and single/dual magnetic switch assemblies.
- A torque table (Table 2) is available for updated joints on the assemblies.
- Figure 4 shows the wiring terminals that **must** be used to wire the starting motor correctly. The T50C (Ignition 'S') terminal **must** be balanced using 16AWG wire between both T50C (Ignition 'S') starting motor terminals.

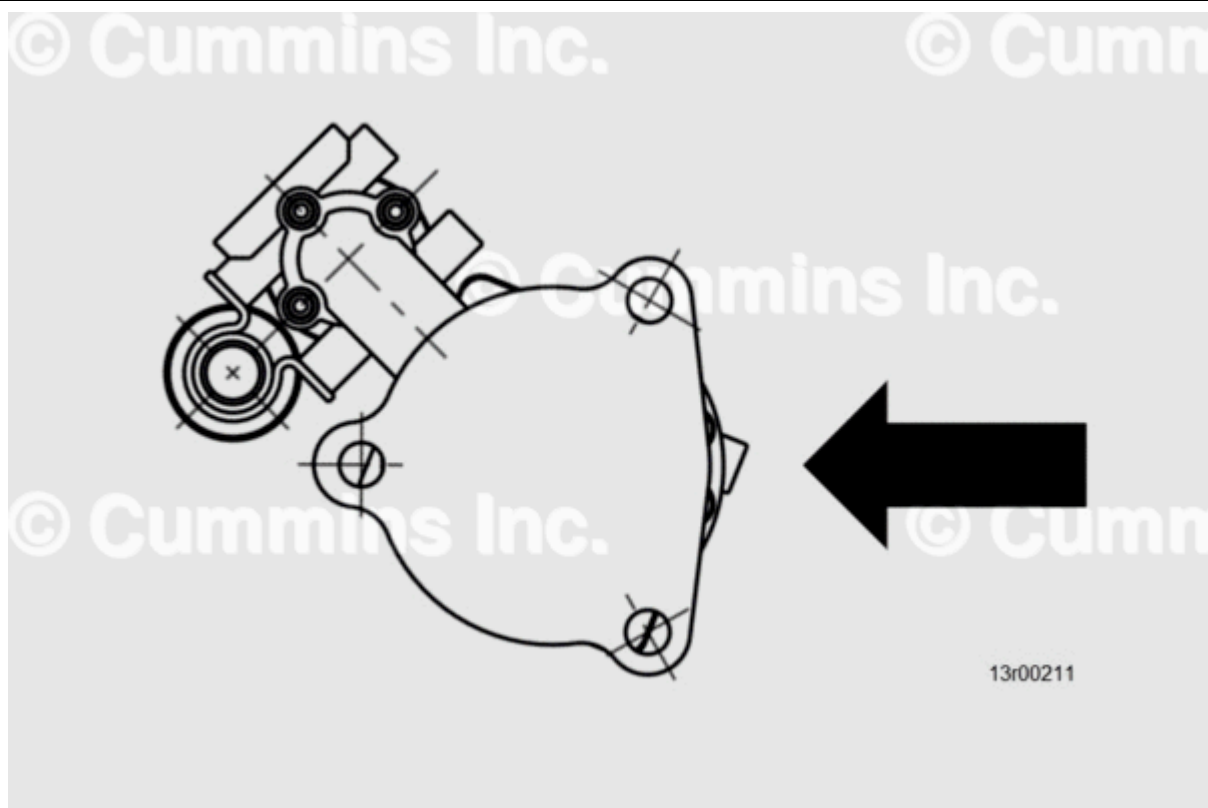


Figure 1, Arrow Indicating the Flat Side of the Starting motor Mounting Flange.

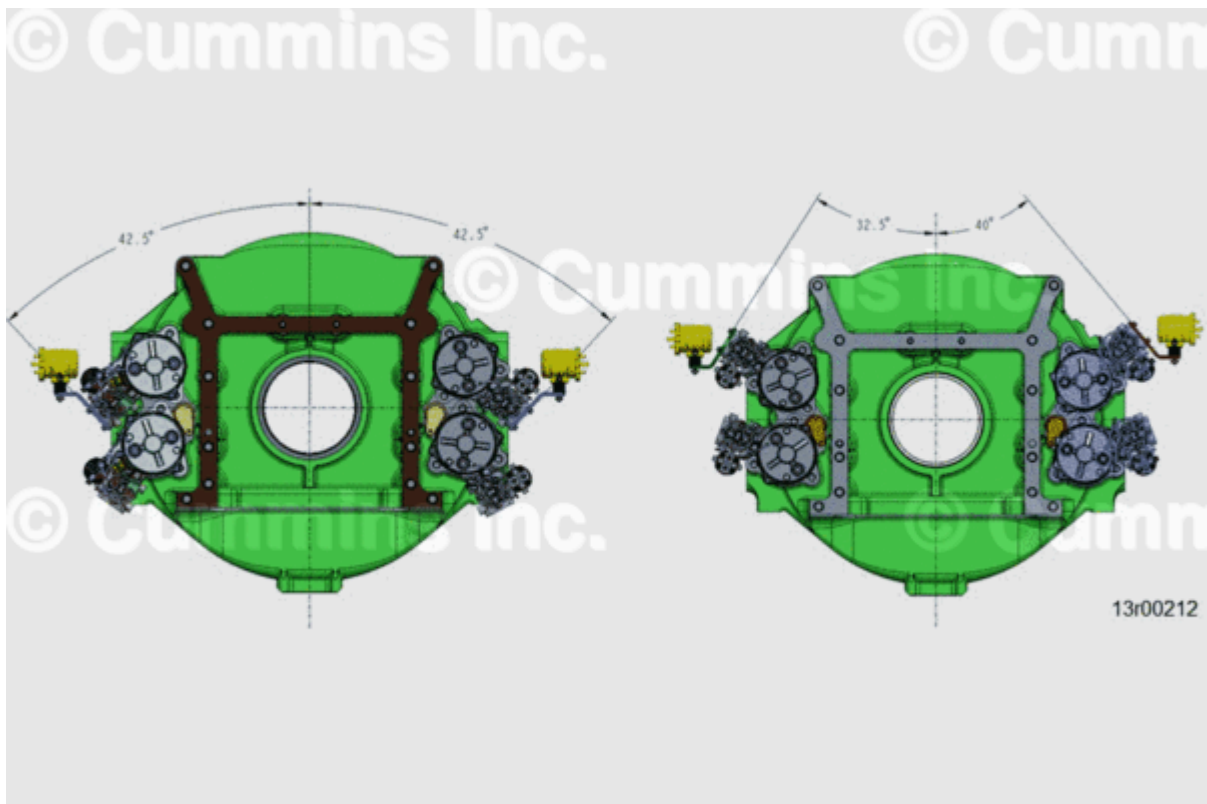


Figure 2, Magnetic Switch Orientations. Left: QSK. Right: KV.

Table 1, Magnetic Switch Bracket Part Numbers

Engine	Engine bank	Dual	Single
KV	Left	6377449	6377447
	Right	6377448	6377446
QSK	Left	6377445	6377435
	Right	6377445	6377435

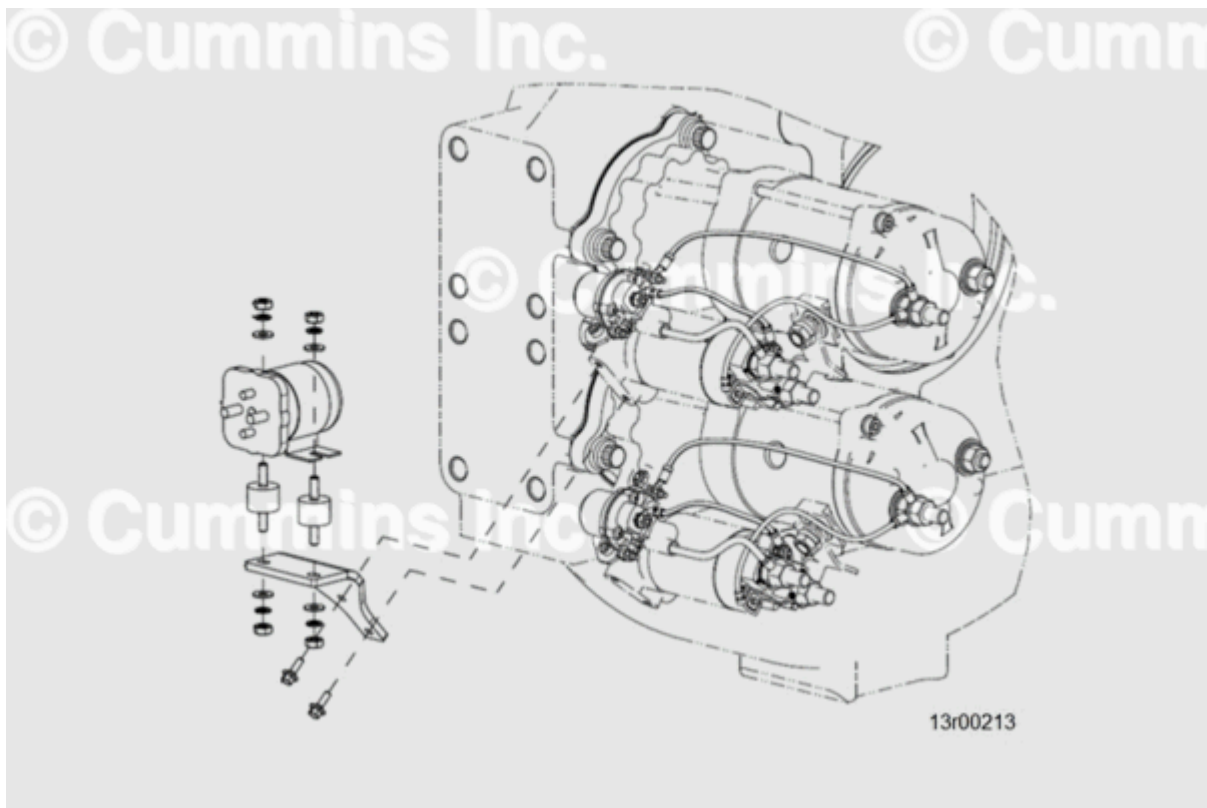


Figure 3, Magnetic Switch Exploded Diagram.

Table 2, Torque

Part Number	Joint Description	Torque (Nm)
3913638	Mounting Bracket	8
-	M5 starting motor Ignition (S/T50C) Terminal	2 to 2.5

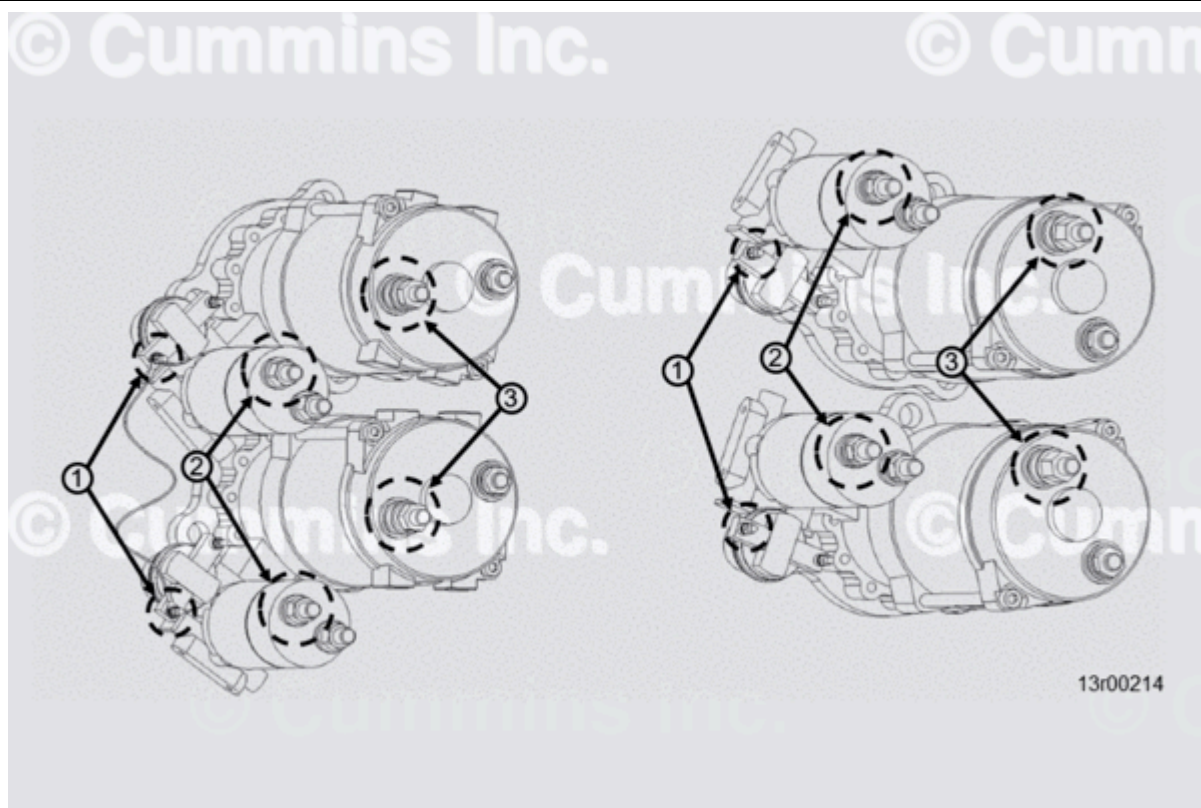


Figure 4, Starting Motor Terminal Identification. Left: QSK. Right: KV.

1. T50C (Ignition 'S') Terminal – **Must** be balanced using 16AWG wire.
2. T30 (B+) Terminal – Outboard most terminal.
3. T31 (GND) Terminal – Longer of the two Terminals.

Service Parts Availability

Service parts are available. See Table 3 for part numbers.

Table 3, Service Parts						
Part Description	KV / QSK	Manufacturer Description	Existing Part Number	Obsolete	Superseded	New Part Number
MOTOR, STARTING	QSK	(MS7)	3632273	Yes	Yes	5541784
			3632274	Yes	Yes	5541783
	KV	(MS7)	3632273	Yes	Yes	5542145
			3632274	Yes	Yes	5538989
	QSK	(M125)	3647165	Yes	Yes	5541784
			3647166	Yes	Yes	5541783
	KV	(M125)	3647165	Yes	Yes	5542145
			3647166	Yes	Yes	5538989
	-	(M128)	5541783	No	No	-
			5541784	No	No	-
			5538989	No	No	-
			5542145	No	No	-
BRACKET, MAGNETIC SWITCH	-	-	6377449	No	No	-
			6377448	No	No	-
			6377445	No	No	-
			6377445	No	No	-
			6377447	No	No	-
			6377446	No	No	-
			6377435	No	No	-
			6377435	No	No	-
SCREW, CAP	-	-	3913638	No	No	-

Part Compatibility

- Full compatibility when used in single starting motor applications.
- No compatibility when used in dual or triple starting motor applications.
- The obsolete MS7 and M125 are hard engagement starting motor. The M128 starting motor is a soft engagement starting motor. Use of hard and soft engagement starting motors together is **not** allowed.

Part Inventory Action

Use existing part number inventory before using the new part number.

Production Status

Implemented for production. See Table 4.

Table 4, Production Information

ESN First

Build Date¹

Plant

33228264

29 April 2022

Daventry

¹Engine build date can be found on the engine dataplate.

Document History

Date	Details
2022-9-21	Module Created

Last Modified: 03-Oct-2022